

1 Quick Review

From here on out you will be breaking into the core of Linear Algebra (and personally, the most fun part!). Everything from here on out will feel very unfamiliar, but very well without your grasp!

In order to make Linear Algebra applicable (both in the real-world and theoretically), we first have to make some assumptions. There is no way we can *prove* these assumptions, we can only really declare them as true. To indicate that these are assumptions on which everything will hinge upon, we call them **axioms**. For vector spaces, a mathematical “terrain” that contains a *set of vectors* (more on what a *set of vectors* is will be covered in sections 4.3–4.4), we must follow 10 axioms (note that the below is a definition, not a theorem!).

Definition 1: Axioms for a Vector Space

Let V be a nonempty set of objects in which *addition* and *scalar multiplication* are defined. *Addition* of two objects is a *sum*, defined by $\mathbf{u} + \mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ are arbitrary objects in V . *Scalar multiplication* is a process by which we associate an object $k\mathbf{u}$ with $\mathbf{u}$ where k is itself not in V but $k\mathbf{u}$ is. For the set V to be a **vector space** and $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ to be vectors, the following axioms **MUST** be satisfied:

1. If $\mathbf{u}, \mathbf{v} \in V$, then $(\mathbf{u} + \mathbf{v}) \in V$.
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. There must exist $\mathbf{0} \in V \mid \mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u}$ ($\mathbf{0}$ is called the *zero vector*)
5. $\forall \mathbf{u} \in V \exists -\mathbf{u} \in V \mid \mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$ ($-\mathbf{u}$ is called the *negative* of $\mathbf{u}$)
6. If k is any scalar, then $k\mathbf{u} \in V$
7. $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$
8. $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$
9. $k(m\mathbf{u}) = (km)(\mathbf{u})$
10. $1\mathbf{u} = \mathbf{u}$

As a side note, I was very intentional in putting as much “mathiness” as possible in the above axioms. It is *very* helpful to know how to read these statements and it will make your proofs shorter to write too! (If you want something a bit easier to read, the textbook has the same definitions on page 184)

The two axioms that are bolded (axioms 1 and 6) are key to showing that some vector space is actually a *subspace* of another, larger vector space (for example, $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$). If you know that a proposed vector space W is part of a larger vector space V **AND** that these axioms hold, you have proven that W is itself a vector space.

There isn’t much more space on this page, but below are two important ideas that will come up very frequently!

Linear Independence: A nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ in a vector space V is linearly independent iff

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_r\mathbf{v}_r = \mathbf{0} \quad \text{only when} \quad k_1 = k_2 = \dots = k_r = 0.$$

Span: The span of a nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is a vector space W consisting of all possible linear combinations of S .

$$W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\} = \text{span}(S)$$

2 Problems

1. Below are proposed definitions for a set of objects S , *vector addition*, and *scalar multiplication*. (i) Write three vectors that would satisfy the proposed condition for S , (ii) determine if they will make up a vector space, and (iii) prove that certain axioms hold.

(a) S is the a set of vectors defined by $S = \{(x^2, y^2) \mid x, y \in \mathbb{R}\}$. Vector addition and multiplication is that of ordinary $\mathbb{R}^n$ vector addition and multiplication.

- i. $(\quad, \quad) (\quad, \quad) (\quad, \quad)$
- ii. Is this a valid vector space?
- iii. **(PROOF)**. Prove whether or not axioms 6 and 7 will hold.

(b) Q is the a set of vectors defined by $Q = \{(xy, 0) \mid x, y \in \mathbb{R}\}$. Vector addition and multiplication is that of ordinary $\mathbb{R}^n$ vector addition and multiplication.

- i. $(\quad, \quad) (\quad, \quad) (\quad, \quad)$
- ii. Is this a valid vector space?
- iii. **(PROOF)**. Prove whether or not axioms 3 and 8 will hold.

- iv. Does Q span $\mathbb{R}^2$?
- v. Is Q a linearly independent set of vectors? Phrased another way, can you get the same vector using two different combinations of x and y ?

2. Since vector spaces need to satisfy only 10 axioms (and if they're a subspace, only 2!), there is a lot of freedom in the definition of a vector space. Write an example vector for each specified vector space.

(a) $\mathbb{P}_4$:

(b) $\mathbb{C}_3(-\infty, \infty)$:

(c) $\mathbb{R}_\infty$:

(d) **(BONUS)**. Going off of the number of values associated with the vector spaces from parts (a), (b), and (c), which one is similar to $\mathbb{R}^5$?

3. Let S be a set with a single vector, meaning $S = \{\mathbf{v}\}$.

(a) **(PROOF)**. Assuming that $\mathbf{v} \neq \mathbf{0}$, prove that S is linearly independent.

(b) Now, let $\mathbf{v} = (k, 2k)$, where $k \in \mathbb{R}$. Does S span $\mathbb{R}_2$? Is S a vector space?

(c) Letting $\mathbf{v}$ be the same as defined in part (b) but with the added restriction of $k \neq 0$, is S a vector space? If so, how do you know? If not, why?

4. Let $S = \{(5x^2 + 3x), (4x + 6), (7)\} \in \mathbb{P}_2$.

(a) Write all of the vectors as a three-tuple (a three-tuple is written as (a, b, c)).

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(b) Show that this set of vectors is *linearly independent*.

(c) $\mathbf{u} = (3x^2 + 6x + 19)$ is a vector chillin' in $\mathbb{P}_2$. Write $\mathbf{u}$ as a linear combination of the vectors in S .
(Though you haven't learned this yet, S is a *basis* for $\mathbb{P}_2$).

(d) **(BONUS)**. In part (a) you converted a vector in $\mathbb{P}_2$ to a vector in $\mathbb{R}^3$. This is called an *isomorphism*.
Based on this, what do you think an *isomorphism* is?

(e) **(BONUS)**. In part (d) you were introduced to the idea of an *isomorphism*. In general, you cannot use an isomorphism if two vector spaces have different *dimensionalities*. Using this information and the common definition of 'dimensions,' what might the *dimensionality* of $\mathbb{P}_2$ and $\mathbb{R}^3$ be?
